

TRAMDif: Texture Retention fashion image editing with Attention enhancement and Mask prediction in Diffusion Model

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Abstract—Text-guided image editing model has demonstrated substantial efficacy in general domain. However, direct application of these models within the fashion domain often leads to three main issues: (1) Inaccurate localization of editing region; (2) Weak editing magnitude; (3) Insufficient texture retention. To address these issues, we propose Texture Retention fashion image editing with Attention enhancement and Mask prediction in Diffusion Model (TRAMDif). Specifically, to obtain more accurate editing regions, the Mask Prediction Module (MPM) is proposed, where the densepose, foreground region and mask prompts from large language model are fed into a lightweight U-Net to predict the mask and a Mask Processor then refine this mask for accurate local editing. To strengthen the editing magnitude, the Attention-Enhanced Diffusion Model is proposed, in which the attention map, noise maps and the mask from Mask Prediction Module are fed into the proposed Attention Processor to produce a refined noise map. By integrating the refined noise map into the diffusion model, the edited image can better align with the target prompt. To tackle the issue of insufficient texture retention, a new Self Value Injection Module (SVIM) is proposed, which incorporates high-frequency information representing texture from the inversion process into the DDIM sampling process. Recognizing the lack of benchmark in fashion image editing, we developed Fashion-E. This new benchmark dataset consists of 2927 unique images, each with several target prompts, specifically curated for four distinct fashion-related editing tasks. Our comprehensive experiments on Fashion-E demonstrate that the proposed method accurately predicts editing regions, substantially boosts editing magnitude, and effectively retain the original texture, outperforming existing state-of-the-art approaches.

Index Terms—Text-guided Image Editing, Fashion Domain, Diffusion Model

I. INTRODUCTION

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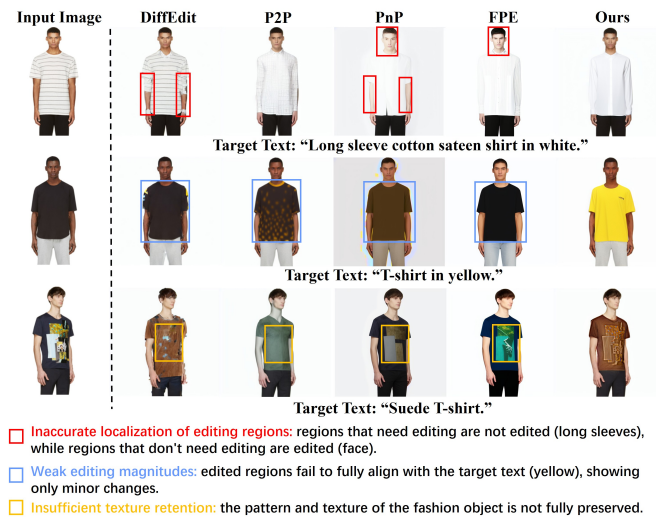


Fig. 1. General-domain text-guided image editing models have inaccurate localization of the editing region (e.g. wrongly handle the sleeve length), weak editing magnitude (e.g. cannot change the color from black to yellow) and insufficient texture retention (e.g. cannot fully preserve the original pattern).

DUE to advances in diffusion models [1] [2] and text-to-image generation models [3]–[5], text-guided image editing has also experienced rapid development. Some models perform training-free image editing through operations like cross-attention and self-attention, such as FPE [6] and P2P [7]. Other models, like DiffEdit [8] and Blended Latent Diffusion [9], focus on localized editing by combining masks. These models have achieved excellent results on various text-guided editing tasks across general domains. In addition to general-domain image editing, image editing tasks in the fashion domain also hold significant research value due to their wide range of applications and high commercial potential [10], [11].

Although text-guided image editing models have demonstrated impressive performance in general domain, their direct application to the fashion domain often encounters several challenges, including: (1) Inaccurate localization of editing regions; (2) Weak editing magnitudes; (3) Insufficient texture retention. As illustrated in the first row of Fig. 1, DiffEdit [8] and PnP [12] fail to perform accurate edits related to sleeve length, while FPE [6] introduces unintended changes to the facial region, highlighting the problem of inaccurate localization. In the second row, the t-shirt edited by general-domain models largely maintains its original black color

rather than reflecting the yellow hue specified in the input text, indicating a weak editing magnitude. In the third row, models such as P2P and PnP fail to preserve the original garment's pattern and texture, thereby revealing the limitation of insufficient texture retention.

To address these challenges, we propose TRAMDiff, a novel model composed of two key components: Mask Prediction Module (MPM), Attention-Enhanced Diffusion Model. Specifically, to tackle the issue of inaccurate localization of editing regions, we introduce the Mask Prediction Module (MPM). In this module, the foreground segmentation, DensePose map, and a mask prompt generated by a large language model are jointly fed into a lightweight U-Net, which first predicts the regions to be edited. Then the Mask Processor is applied to produce more refined editing mask for accurate local editing. To strengthen the editing magnitude, the Attention-Enhanced Diffusion Model is proposed, where pixels with higher attention values are used to replace the original pixels within the editing region through the proposed Attention Processor. To tackle the issue of insufficient texture retention, The Self Value Injection Module (SVIM) is applied in Attention-Enhanced Diffusion Model, fully incorporating high-frequency information from the inversion process into the sampling process. As shown in Fig. 1, compared with state-of-the-art editing models, our method achieves more accurate localization of the editing regions, stronger editing magnitude, and superior preservation of the original patterns and textures.

Given the absence of benchmarks in fashion image editing, we construct Fashion-E dataset which is composed of 2927 unique images from Fashion-Gen [13] dataset with the carefully annotated target prompts. In this dataset, four editorial tasks in the fashion domain are included: color editing, detail editing, material editing, and comprehensive editing, in order to more comprehensively evaluate the performance of text-guided fashion image editing methods.

We conduct extensive experiments on Fashion-E and evaluate our model using a range of metrics, such as CLIP-T, CLIP-I, SSIM and LPIPS. The experimental results demonstrate that our model can accurately predict the regions that need to be edited, effectively enhance editing strength, and significantly improve the preservation of original textures and patterns in the edited images. Additionally, we conduct an ablation study to evaluate the individual contributions of each component in our proposed method. The main contributions of this work can be summarized as follows:

- Mask Prediction Module (MPM) is proposed to accurately predict the editing region, significantly improving the model's ability to perform precise local edits.
- Attention-Enhanced Diffusion Model is proposed to address the issue of weak editing magnitude in text-guided fashion image editing, enabling more effective and targeted modifications in alignment with the input prompt.
- Novel Self Value Injection Module (SVIM) is proposed and applied in the Attention-Enhanced Diffusion Model, which enable better retention of the original patterns and textures.
- Lacking standardized benchmarks in fashion image editing, we propose Fashion-E, a novel dataset that enables

the assessment of diverse models for various fashion editing tasks.

- Experiments on Fashion-E show our model surpasses state-of-the-art models in text alignment and original information preservation.

Compared to the conference version [14], the main differences in this paper are as follows:

- Instead of the MaskNet used in the conference version, this paper proposes the MPM, which, by combining different editing modes, predicts the editing regions more accurately.
- The SVIM is proposed and applied within the Attention-Enhanced Diffusion Model, significantly enhancing the retention of original texture and pattern information in the edited images.

II. RELATED WORK

A. Text-guided Image Editing

Recent works widely applied diffusion model in text-guided image editing. SDEdit [15] first achieved text-guided image editing through forward noise addition and reverse denoising. Similarly, DiffusionAutoencoder [16] integrates SDEdit's noise addition process with DDIM, proposing DDIM Inversion for editing the semantic information in the image. Based on the DDIM Inversion framework, many works have emerged that applied prompts optimization to enhance the inversion performance, such as Null-text Inversion [17], PRedItOR [18] and Prompt Tuning Inversion [19]. Another methods optimized the calculations of ODEs to improve the performance and efficiency of inversion, such as LEDits++ [20], and ProxEdit [21]. In addition to inversion-based models, some models also perform text-guided image editing through attention modification. Prompt-to-Prompt [22] explored the impact of cross-attention on image generation and, for the first time, performed various editing tasks by applying different operations to the cross-attention maps. Plug-and-play [23] and FPE [6] explored the impact of self-attention on image editing and improved the spatial consistency between the edited image and the original image by replacing the self-attention map. Masactrl [24] achieves highly consistent object editing by replacing the key and value matrices in the self-attention.

However, both inversion-based and attention-based editing methods face the issue of insufficient editing magnitudes. This is mainly because these methods use generative models (e.g., Stable Diffusion) rather than models specifically designed for editing as the baseline. Therefore, some instruction-based models specifically designed for editing have been proposed to address this issue. Instruct-Pix2Pix [25] leveraged Prompt-to-Prompt [22] and GPT-3 [26] to generate a large amount of samples consisting of original images, instructions and edited images. These samples are then used to fine-tune Stable Diffusion, enabling it to directly edit images through instructions. MagicBrush [27] fine-tuned Instruct-Pix2Pix with 10k high-quality real editing data, improving the editing intensity, accuracy, and aesthetics. HIVE [28] incorporates human preferences into instruction-based models by training a reward model and using an improved reinforcement learning strategy.

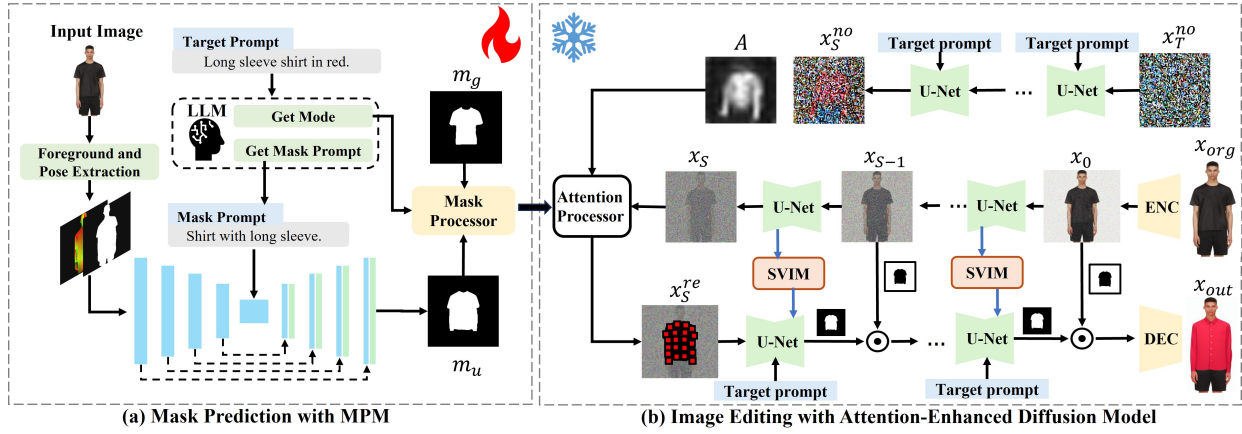


Fig. 2. Overview of our model. (a) Mask Prediction Module: The input image and target prompt are preprocessed to get the foreground region, densepose map and mask prompt, which are then input into a lightweight U-Net to predict the editing region. Then the Mask Processor is applied to produce more refined editing mask for accurate local editing. (b) Attention-Enhanced Diffusion Model: DDIM inversion and sampling are conducted to get the attention and noise maps. Then all these maps and the masks from Mask Prediction Module are fed into the Attention Processor to create a refined inversion map. The refined inversion map is employed as the initialization for a subsequent DDIM sampling, during which the high-frequency information from SVIM is progressively incorporated into the denoised latent.

This enables the images edited by HIVE to better align with human preferences.

Although instruction-based image editing methods have strong editing capabilities, experiments have shown that they often suffer from inaccurate localization of the editing area, failing to ensure that only the regions relevant to the text are altered. In summary, the existing models (inversion-based, attention-based, and instruction-based) do not adequately meet the demands of the fashion domain.

B. Local Image Editing

Local editing aims to ensure that the editing occurs only within specified regions, while keeping the content outside those regions as unchanged as possible. Stable Diffusion [4], GLIDE [29] and SD-ControlNet [30] trained an inpainting version of the model to restrict the editing to the masked areas. To allow the content within the mask to change more smoothly, Blended Latent Diffusion [9] mixed noisy versions of the original image with the intermediate results of the text-guided diffusion to ensure consistency between the masked area and the text. However, The methods mentioned above all require the user to manually specify the areas that need to be edited. To automate local editing, DiffEdit [31] obtains the mask of the editing region by comparing the differences between the latents of two generative pipelines and send the mask to blended latent diffusion [9] to complete editing. Thanks to the powerful zero-shot semantic segmentation capability of Segment Anything [32], Grounded-SAM [33] combines it with an open-vocabulary object detection model [34] to create a model that can segment the corresponding masked regions of images based on text prompts. Grounded-SAM [33] and SAM [32] has since been widely applied to various local editing models, such as Grounded Instruct-Pix2Pix [35], ZONE [36].

Although the above models can, to some extent, identify the regions that need to be edited, in the fashion domain, many tasks require not only segmenting a mask based on the original image but also predicting the regions that need to be edited

based on text (e.g., editing a short-sleeve into a long-sleeve). Therefore, these models still encounter the issue of inaccurate localization of the editing regions.

C. Text-guided Fashion Image Editing

FICE [37] optimized the image in the latent space of StyleGAN [38] through losses according to text, pose, and masks, which can edit clothing into various different styles. TexFit [39] trained a lightweight network for segmenting the editing region and a diffusion model for fashion editing based on the mask. MGD [40] and Ti-MGD [41] used garment sketches and human pose maps as additional conditions to more precisely control the regions that need to be edited. FashionMatrix [42] integrated large language models into fashion editing, achieving a preliminary interactive fashion editing agent. In addition, some classic general-domain models, such as Stable Diffusion Inpainting [4], ControlNet [30], and BLIP-2 [43], frequently appear in the fashion domain.

Although these existing models have made some improvements for the fashion domain, they still face two main issues: First, the editing regions are either simply segmented (TexFit) or difficult to obtain (MGD); Second, they lack effective methods for preserving the original textures and details. In contrast, our proposed TRAMDiff model not only ensures the fashion editing magnitude but also accurately predicts the regions that need to be edited based on text. Additionally, it significantly preserves the original textures and pattern information.

III. METHOD

A. Model Overview

Our method consists of two main components: Mask Prediction Module (MPM) and Attention-Enhanced Diffusion Model, as shown in Fig. 2. In the Mask Prediction Module, input image and target prompt are first processed by Graphonomy [44], DensePose [45] and LLAMA3-8b to get

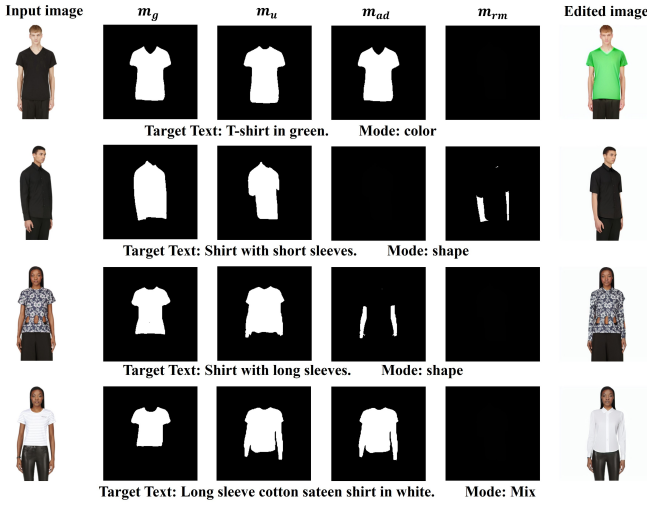


Fig. 3. Comparison of different mask strategies under different modes. m_{ad} is the area that needs to be added and m_{rm} is area that needs to be removed.

the foreground region, densepose map and mask prompt, which are input into a lightweight U-Net to predict the mask of editing region. Then, the mask predicted by the U-Net, the cloth mask segmented by Graphonomy, and the mode predicted by the LLM will be input together into the Mask Processor to accurately produce a mask that represents the region to be modified and a mask that represents the region to be removed. In the Attention-Enhanced Diffusion Model, DDIM inversion from input image and DDIM sampling from random noise are first conducted to obtain the attention map, noise map and inversion map. Then the attention map, noise map, inversion map and the masks from Mask Prediction Module, are input into the Attention Processor to produce a refined inversion map. Finally, the refined inversion map is employed as the initialization for a subsequent DDIM sampling. During this process, the SVIM is utilized to progressively incorporate original high-frequency information into the denoised latent, thereby preserving the original patterns and textures. Concurrently, the region needs to be edited and the region that does not need to be edited are also blended to ensure that the regions unrelated to the target prompt remain unaffected.

B. Mask Prediction Module

The Mask Prediction Module is introduced to precisely identify the editing region, guided by the target prompt and input image. As shown in Fig. 2(a), its primary network is a lightweight U-Net, where spatial attention layers are only used in the middle block to integrate textual information. To enable the U-Net to predict the areas that need to be edited rather than merely segmenting existing regions, we concatenate the foreground mask and DensePose map as model input instead of the original image. The foreground mask and DensePose map are produced by the Graphonomy model and DensePose model, respectively. Furthermore, since the editing region is solely determined by shape-related terms in the target prompt, LLAMA3-8b processes the prompt to generate a mask prompt that only includes words describing the fashion object's shape.

During the training of the U-Net, the matched image-text pairs along with their corresponding masks are prepared, and the MSE loss between the predicted mask and the cloth mask is applied as the optimization objective.

In fashion editing tasks, the mask standards vary depending on the specific task. For example, in color editing tasks, only the mask of the original fashion item needs to be segmented, whereas for shape-related editing tasks, it is necessary to predict which areas should be added and which should be removed. Therefore, to further improve the accuracy of local editing, we first categorize the image editing into three modes: color-only editing mode, shape-only editing mode, and color-shape-mix editing mode, which can be predicted by LLAMA3-8b. Additionally, the mask is also divided into two specific regions: m_{ad} for area that needs to be added and m_{rm} for area that needs to be removed.

Assuming that the cloth mask segmented by Graphonomy is denoted as m_g and the mask predicted by the lightweight U-Net is denoted as m_u , different masking strategies are designed in different editing modes to ensure the accuracy of local edits. As shown in the first line of Fig. 3, for color-only editing tasks, where no shape alteration is involved, we directly set the addition mask m_{ad} as the cloth mask m_g and the removal mask m_{rm} as empty set. As shown in the second and third lines of Fig. 3, for shape-only editing tasks, which require precise modifications to local details (e.g., adjustments to the sleeve length), we define the addition region m_{ad} as the area included in m_u but excluded from m_g , and the removal region m_{rm} as the area included in m_g but excluded from m_u . As shown in the last line of Fig. 3, for mix editing tasks, which involve changes in both the overall color and local structure of the fashion item, we set m_{ad} to m_u and define m_{rm} as the area included in m_g but excluded from m_u . The specific computation of m_{ad} and m_{rm} is performed by the Mask Processor, as detailed in Equation 1:

$$\begin{cases} m_{ad} = m_g, \\ m_{rm} = \emptyset & \text{if mode = color} \\ m_{ad} = m_u \wedge (1 - m_g), \\ m_{rm} = (1 - m_u) \wedge m_g & \text{if mode = shape} \\ m_{ad} = m_u, \\ m_{rm} = (1 - m_u) \wedge m_g & \text{if mode = mix} \end{cases} \quad (1)$$

where $\wedge$ denotes the logical AND operation. m_{ad} and m_{rm} are the final outputs of MPM and will be used as the inputs to the Attention Processor.

C. Attention-Enhanced Diffusion Model

As depicted in Fig. 1, many existing editing methods struggle to generate images consistent with the target prompt when significant visual changes are required, such as transforming a black t-shirt to yellow. To investigate the causes of these editing failures, we employ a 50-step DDIM Inversion + DDIM Sampling process for two image editing tasks—editing 200 black T-shirts to red or yellow—and compare the text-image cross attention maps across each task. The method for

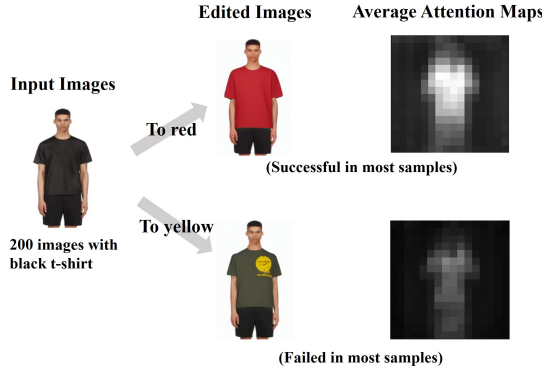


Fig. 4. A comparison of attention maps between successful and failed fashion editing cases is conducted. Specifically, 200 black t-shirts are edited to red and yellow, respectively, and the average attention maps corresponding to each case are computed to analyze the relationship between attention value and editing success.

obtaining the text-image cross attention maps is as follows: we first average the text-image cross attention maps from all layers of the UNet at 16×16 resolution across 50 steps along the text dimension to obtain the text-image cross attention map for each step, then compute their average to derive the final text-image cross attention map. As shown in Fig. 4, successful editing (i.e. editing black T-shirts to red) correlates with high values in the text-image cross attention map, and vice versa. Consequently, our method boosts editing magnitude by increasing these attention map values.

1) *Model Overview*: Built upon the Stable Diffusion backbone, the Attention-Enhanced Diffusion Model operates in a training-free manner. The editing procedure commences with the encoding of the input image x_{org} into its latent representation, x_0 . Following this, DDIM Inversion is initiated, using x_0 as its starting point:

$$x_{t+1} = \sqrt{1 - \alpha_{t+1}}\epsilon_{\theta}(x_t, t) + \sqrt{\alpha_{t+1}}f_{\theta}(x_t, t), \quad (2)$$

where t is the time step, α_t is a coefficient that decreases over time steps, ϵ_{θ} is the noise prediction from the denoising U-Net and f_{θ} is calculated by $(x_t - \sqrt{1 - \alpha_t}\epsilon_{\theta})/\sqrt{\alpha_t}$. The complete inversion procedure spans S steps, transforming x_0 into x_S , a noise map that encapsulates the spatial characteristics of the input image. Throughout this process, both the DDIM trajectory $x_0, x_1, \dots, x_{S-1}$ and the resulting inversion noise map x_S are meticulously collected.

Concurrently, DDIM sampling is also performed, commencing from a random noise x_T^{no} that shares the same dimensions as x_0 :

$$x_{t-1}^{no} = \sqrt{\frac{\alpha_{t-1}}{\alpha_t}} \left(x_t^{no} - \frac{\alpha_{t-1} - \alpha_t}{\alpha_{t-1}\sqrt{1 - \alpha_t}} \epsilon_{\theta}(x_t^{no}, t, c) \right), \quad (3)$$

where α_t is the same coefficient as in DDIM Inversion, c is the target prompt, and ϵ_{θ} represents the noise prediction from the denoising U-Net. After $(T - S)$ steps of DDIM sampling, x_T^{no} is transformed into x_S^{no} through denoising. The attention map A is then determined by averaging all cross attention maps with a resolution of 16×16 within the U-Net.

Following the DDIM Inversion and sampling steps, the noise map x_S^{no} , its corresponding attention map A , the inver-

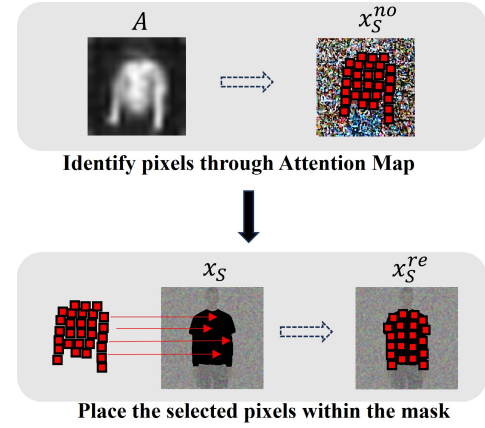


Fig. 5. Details of Attention Processor. The pixels with higher attention values are identified and then placed within the predicted editing region.

sion noise map x_S , and the masks m_{ad} and m_{rm} from Mask Prediction Module are all fed into the attention processor. This process yields a refined noise map, x_S^{re} , which exhibits a higher attention map value.

Finally, DDIM sampling is performed using the refined noise map x_S^{re} , guided by the target prompt. This sampling incorporates the SVIM to inject original high-frequency information into the denoising steps, and the resulting intermediate outputs are then blended with the DDIM trajectory [9].

$$x_{t-1}^{re} = DDIM(x_t^{re}) \odot m + x_{t-1} \odot (1 - m), \quad (4)$$

where x_t^{re} signifies the refined noise map at time step t , $m = m_{ad} \vee m_{rm}$ indicates the precise region to be edited. Executing Equation (4) for S steps allows x_S^{re} to be denoised, yielding x_0^{re} . The edited image x_{out} is subsequently generated by feeding x_0^{re} into the decoder.

2) *Attention Processor*: The procedures of Attention Processor are illustrated in Fig. 5. Since editing for addition and modification differs significantly from editing for removal, with the former requiring an increase in attention values for the corresponding region and the latter requiring a decrease, we adopt different processing strategies for the m_{ad} and m_{rm} regions in the Attention Processor.

Specifically, given an addition mask m_{ad} that encompasses N pixels, the pixels within the region defined by m_{ad} in map x_S can be expressed as:

$$G_{ad} = \{pix | m_{ad}(pix) > 0, pix \in x_S\}, \quad (5)$$

where pix is the pixel in noise map and G_{ad} represents the set of pixels under the mask m_{ad} in the inversion noise map x_S . Equation (5) thus specifies the regions where changes in both appearance and shape are required.

The Attention Processor then proceeds to identify the $N/2$ pixels exhibiting the highest attention map values in x_S^{no} :

$$G_{pr} = \{pix | A(pix) > V_{min}, pix \in x_S^{no}\}, \quad (6)$$

where G_{pr} is defined as the set of pixels with greater attention values derived from x_S^{no} and V_{min} signifies the $(0.5N)$ th largest value within the attention map A .

Finally, all pixels within G_{ad} are orderly replaced by pixels from G_{pr} . In cases where all pixels in G_{pr} have been used, the

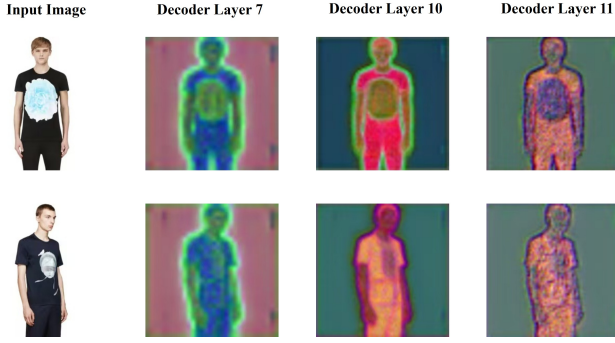


Fig. 6. PCA visualization of value matrix at the last inversion step.

remaining pixels in G_{ad} are then filled by randomly selected ones from G_{pr} .

It's worth noting that the resolution of attention map A is 16×16 , while the resolution of noise map x_S^{no} and x_S is 64×64 in stable diffusion [4]. Therefore, attention map A is interpolated to 64×64 to address the resolution discrepancy.

Given the mask m_{rm} , the regions that need to be removed can be denoted as:

$$G_{rm} = \{pix | m_{rm}(pix) > 0, pix \in x_S\}, \quad (7)$$

where pix represents the pixel of noise map and G_{rm} is a set of pixels within the mask m_{rm} in the inversion noise map x_S . The pix in G_{rm} are finally replaced by random noise pixel $\epsilon \in N(0, I)$ to remove this portion of the fashion object, and this process is performed only within the Attention Processor Module.

D. Self Value Injection Module

Mask-based diffusion models, such as Stable Diffusion Inpainting and DiffEdit, often struggle to preserve details within the masked area, such as clothing patterns. The same issue also occurs in the MADiff [14], as it directly replaces the pixels of the inversion noise map. To address this issue, we propose the Self Value Injection Module, which fully incorporates high-frequency information from the inversion process into the sampling process, as shown in Fig. 2.

Inspired by MasaCtrl [24] which preserves fine-grained object details by replacing the key and value matrices from DDIM inversion with those originally used in DDIM sampling, we hypothesize that these key and value matrices contain high-frequency information such as textures. Since the key matrix is involved in dot-product operations, high-frequency information may be lost during this process; thus, we design the Self Value Injection Module based on the value matrix to capture high-frequency information. To verify whether the value matrix indeed contains high-frequency components representing textures, we adopt the PCA-based visualization method from PnP [12] to visualize the value matrices during DDIM inversion, as shown in Fig. 6. The results demonstrate that the value matrix effectively preserves the spatial information of the original image, including high-frequency details corresponding to textures. Therefore, we explicitly extract high-frequency information from the value matrix during DDIM inversion and fuse it into the value matrix during DDIM sampling, thereby

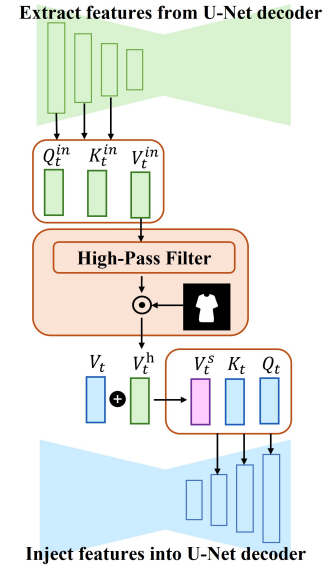


Fig. 7. Details of the proposed Self Value Injection Module (SVIM).

better preserving high-frequency information such as textures. Notably, unlike MasaCtrl, which directly replaces the key and value matrices, our approach explicitly extracts high-frequency information from the value matrix during DDIM inversion, allowing the model to not only perform edits according to task requirements but also retain detailed information such as textures in the original clothing.

The details of Self Value Injection Module is presented in Fig. 7. Specifically, the value matrix from the self-attention layer of the U-Net decoder at each timestep are firstly collected during the DDIM inversion process, denoted as V_t^{in} . Let b , h , w represent the batch size, height, and width of the input feature map, respectively and d denote the dimension of the self-attention layer. Then the V_t^{in} matrix is reshaped from $b \times (hw) \times d$ to $b \times d \times h \times w$ and input into the Self Value Injection Module to get the high-frequency component:

$$V_t^h = (V_t^{in} - G_\sigma * V_t^{in}) \odot m, \quad (8)$$

where V_t^h is high-pass filtered value matrix, $*$ represents convolution, G_σ is Gaussian blur kernel with standard deviation σ , and $m = m_{ad} \cup m_{rm}$ is the union of m_{ad} and m_{rm} .

The high-pass filtered value matrix V_t^h is added to the corresponding value matrix V_t with the same timestep and position during the DDIM sampling process:

$$V_t^s = V_t + V_t^h, \quad (9)$$

where V_t^s is the value matrix with more original high-frequency information. By replacing V_t with V_t^s during the DDIM sampling process, the edited image retains more of the original image information, as shown in Fig. 7.

IV. EXPERIMENTS

A. Dataset and Metrics

To train the Mask Prediction Module (MPM), we collect a subset from FashionGen dataset, comprising 28,390 samples within nine different fashion categories (shirts, tops, sweaters,



Fig. 8. Fashion-E dataset. The input images and corresponding target texts for the four fashion editing tasks are presented, serving as the basis for evaluating the performance of editing models.

jeans, pants, skirts, dresses, shorts, hats). Apart from containing the text description from FashionGen, each sample is also annotated with text-related mask as the ground truth for training the MPM.

Given the limitations of existing fashion datasets in comprehensively assessing models for various fashion tasks (such as color, shape, and material manipulation), we construct a benchmark dataset named Fashion-E. This benchmark comprises 2927 unique fashion images, each with several target prompts, and is structured to facilitate evaluation across four distinct fashion editing categories: color, detail, material, and comprehensive editing. The overview of the Fashion-E benchmark are shown in Fig. 8 and Table I. In color editing task, two types of sub-tasks are considered: (1) editing with significant visual differences and (2) regular color or textures editing. Consequently, around 300 dark-colored garments and light-colored garments are selected from Fashion-Gen dataset. For the editing with significant visual differences, the dark-colored garments are required to be edited to white/yellow while the light-colored garments are required to be edited to black/red. For the regular color or textures editing, both the dark-colored and light-colored garments are required to be edited to blue/green/orange/purple/camouflage. Detail editing involves six specific tasks distributed across 4 sub-sets: transitioning from round-collar to v-neck/spread-collar, from spread-collar to v-neck/round-collar, from long-sleeve to short-sleeve, and from short-sleeve to long-sleeve. Each sub-set comprises approximately 350 fashion images accompanied by corresponding masks tailored for mask-based image editing models. In addition, we have also included about 300 editing samples featuring shoes and accessories to enhance category diversity in the benchmark. For material editing, the sub-dataset encompasses approximately 450 fashion images. To ensure diversity within the dataset, each image prominently exhibits notable variations in clothing form, color, and poses. Five common materials are selected from the text descriptions of Fashion-Gen dataset as the editing targets, namely: denim, suede, lace, canvas and mesh, representing prevalent fabric types within the fashion domain. Lastly, in the comprehensive editing task, about 450 fashion images are selected with the

TABLE I
STATISTICAL ANALYSIS OF THE FASHION-E DATASET.

Task	Editing targets	Unique images	Number of samples
Color	9 different colors	306	2142
Detail	garment shapes / shoes / accessories	1706	2296
Material	5 different materials	447	2235
Comprehensive	mixture of different editing tasks	468	2340
All	-	2927	9013

principle of diversity. Five substantially distinct lengthy texts, including various cloth attributes such as sleeve length, color, and category, are sourced from the Fashion-Gen dataset to serve as editing targets. This task aims to evaluate the model's proficiency in handling complex editing scenarios involving multifaceted garment attributes. Each sample in this set is paired with a target text description specifying a particular editing task.

For evaluation, we adopt the following metrics: CLIP Text Score (CLIP-T) [46] to measure the alignment between the edited image and the target text, CLIP Image Score (CLIP-I) [46] to assess the preservation of original image content, SSIM [47] to evaluate the structural similarity between the original and edited images, and LPIPS [48] to measure perceptual similarity from a human visual perspective.

B. Comparison with State-of-the-Art Methods

The proposed method is evaluated against nine state-of-the-art text-guided image editing models on the Fashion-E benchmark: DiffEdit [8], FICE [37], P2P [7], PnP [12], IP2P [25], FPE [6], LEDITS++ [20], HIVE [28], and the conference version of our approach, MADiff [14]. All experiments are conducted using the official implementations.

1) *Qualitative Comparisons*: Visualization results of different models are shown in Fig. 9. PnP and IP2P exhibited significant inaccuracies in localizing the editing regions, often making additional erroneous edits to areas unrelated to the target prompt, especially the background of the original image. P2P and FPE demonstrate relatively accurate localization of the editing regions, but their editing magnitudes are insufficient. As shown in the second and sixth rows of Fig. 9, the edited images do not align well with the target prompt. In addition, they also face the issue of insufficient consistency in textures and patterns on the clothing before and after editing. LEDITS++ exhibits sufficient editing strength, but its localization of editing regions is inaccurate. It is particularly prone to altering facial information in the original image, which compromises the identity of the person. Moreover, its edits tend to be overly "global", preserving only the rough structure or texture of the original image while losing many high-dimensional details, as shown in the fourth and seventh rows of Fig. 9. HIVE handles color and material editing tasks well and is able to preserve the original clothing textures effectively in color editing tasks. However, it often produces suboptimal results when dealing with shape-changing tasks, such as detail and comprehensive edits. Additionally, similar to IP2P, its localization of editing regions is not sufficiently accurate. The conference version of our method, MADiff is able to accurately localize the regions that need to be edited

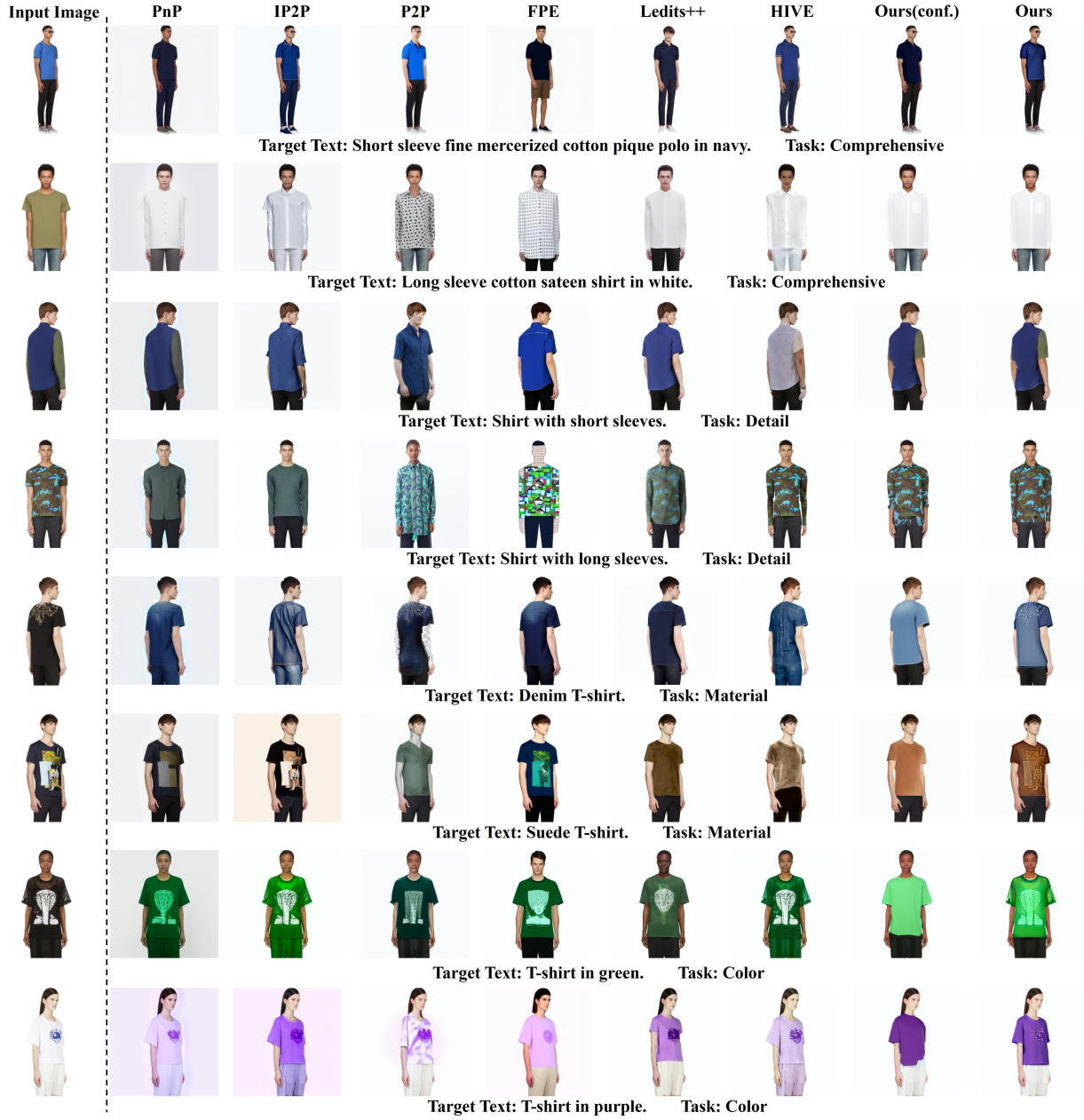


Fig. 9. Qualitative comparison with State-of-the-Art methods on Fashion-E. The target texts are placed below the row of images.

and demonstrates strong editing strength. However, due to its direct replacement of pixels in the inversion map, it fails to preserve the original texture and pattern information, as shown in rows five to eight of Fig. 9. However, our method, TRAMDiff, shows sufficient editing magnitude, accurate localization of editing regions and great retention of the original texture and pattern. This indicates that our model is better suited to image editing tasks in the fashion domain.

2) *Quantitative Comparisons:* The quantitative experiment is first conducted on the proposed Fashion-E benchmark. We use multiple metrics to comprehensively evaluate the model's editing magnitudes, editing region localization accuracy, and retention of pattern and texture information. As presented in TABLE II, in terms of the CLIP-T score, our model far outperforms text-guided image editing models (such as

DiffEdit, FPE, etc.), and even surpasses instruction-based image-guided models that are specifically trained on image editing datasets (such as instruct-Pix2Pix, HIVE). This demonstrates that our approach significantly enhances the model's editing magnitude, as the images edited with our method align well with the target prompt. Additionally, on both the CLIP-I, SSIM and LPIPS metrics, our model achieved the best results, significantly outperforming existing models, including our proposed MADiff model. This indicates that the images edited using our method exhibit stronger similarity to the original images. Combining the CLIP-T score with the quantitative results, it is evident that the improvements in these image similarity-related metrics is mainly due to: (1) Our model's ability to more accurately predict the regions that need to be edited while keeping the irrelevant areas unchanged; (2) Our

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON FASHION-E. BOLD FONT INDICATES THE BEST VALUE FOR THE METRIC, WHILE UNDERLINED FONT INDICATES THE SECOND-BEST VALUE.

Methods	Pub./Year	CLIP-T ($\uparrow$)	CLIP-I ($\uparrow$)	LPIPS ($\downarrow$)	SSIM ($\uparrow$)
DiffEdit [8]	ICLR ₂₃	27.01	83.35	0.154	0.785
FICE [37]	Arxiv ₂₃	27.81	81.62	0.453	0.703
P2P [7]	ICLR ₂₃	28.06	85.82	0.146	0.837
PnP [12]	CVPR ₂₃	28.23	87.57	0.146	0.860
IP2P [25]	CVPR ₂₃	29.07	87.76	0.172	0.865
FPE [6]	CVPR ₂₄	27.63	84.09	0.159	0.847
LEDits++ [20]	CVPR ₂₄	28.88	86.77	0.129	0.890
HIVE [28]	CVPR ₂₄	28.23	89.49	0.138	0.895
TexFit [39]	AAAI ₂₄	28.43	<u>90.14</u>	0.122	0.867
Ours(conf.) [14]	ICASSP ₂₅	29.47	89.92	0.139	0.874
Ours	-	<u>29.36</u>	91.40	<u>0.126</u>	0.895

TABLE III
THE METRIC RESULTS FOR SPECIFIC TASKS ON THE FASHION-E BENCHMARK. BOLD FONT INDICATES THE BEST VALUE FOR THE METRIC, WHILE UNDERLINED FONT INDICATES THE SECOND-BEST VALUE.

Method	Color				Detail				Material				Comprehensive			
	CLIP-T	CLIP-I	LPIPS	SSIM	CLIP-T	CLIP-I	LPIPS	SSIM	CLIP-T	CLIP-I	LPIPS	SSIM	CLIP-T	CLIP-I	LPIPS	SSIM
DiffEdit [8]	26.17	82.64	0.149	0.783	25.52	86.28	0.112	0.792	26.30	83.26	0.145	0.810	30.06	81.20	0.210	0.753
FICE [37]	27.09	81.96	0.471	0.688	27.26	81.65	0.447	0.729	28.26	82.24	0.483	0.662	28.62	80.63	0.409	0.733
P2P [7]	27.48	85.62	0.157	0.832	26.44	85.89	0.158	0.829	<u>28.51</u>	85.48	0.131	0.845	29.82	86.28	0.139	0.841
PnP [12]	27.81	87.28	0.156	0.856	26.96	88.08	0.132	0.878	27.18	89.64	0.127	0.873	30.98	85.27	0.170	0.834
IP2P [25]	<u>29.39</u>	89.33	0.188	0.882	27.65	89.71	0.146	0.876	27.66	87.33	0.128	0.872	31.57	84.66	0.224	0.829
FPE [6]	27.35	83.19	0.156	0.878	26.84	84.58	0.157	0.862	26.60	86.45	0.132	0.841	29.74	82.13	0.192	0.805
LEDits++ [20]	28.64	87.46	0.102	<u>0.893</u>	<u>27.65</u>	87.77	0.116	0.880	27.79	88.59	0.129	0.889	31.44	83.26	0.169	0.896
HIVE [28]	28.15	92.72	0.127	0.914	26.97	90.33	0.131	<u>0.882</u>	27.63	86.72	0.148	0.889	30.17	<u>88.17</u>	0.145	0.896
TexFit [39]	29.22	90.53	<u>0.114</u>	0.868	27.49	90.82	0.119	0.848	27.71	89.12	0.112	0.872	29.30	90.09	0.141	0.880
Ours(conf.) [14]	29.68	90.51	0.150	0.868	27.82	<u>91.03</u>	0.133	0.872	28.25	90.90	<u>0.115</u>	<u>0.892</u>	32.14	88.13	0.158	0.863
Ours	29.27	<u>91.26</u>	0.145	0.888	27.46	91.61	<u>0.117</u>	0.900	29.04	92.31	0.101	0.907	<u>31.67</u>	90.43	0.139	<u>0.885</u>

TABLE IV
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON DFMM-SPOTLIGHT DATASET. BOLD FONT INDICATES THE BEST VALUE FOR THE METRIC.

Methods	CLIP-T ($\uparrow$)	CLIP-I ($\uparrow$)	LPIPS ($\downarrow$)	SSIM ($\uparrow$)
IP2P [25]	30.02	89.91	0.136	0.824
LEDits++ [20]	31.34	81.29	0.311	0.673
HIVE [28]	29.44	92.09	0.108	0.860
TexFit [39]	28.10	96.93	0.052	0.940
Ours(conf.) [14]	28.45	96.23	0.048	0.938
Ours	28.43	97.17	0.037	0.948

model's superior capability in preserving the original patterns and textures.

To better evaluate the model's robustness and task-specific generalization, the CLIP-T, CLIP-I, LPIPS, and SSIM metrics for each individual task on Fashion-E are presented in Table III. As shown in the table, our method performs well across all four editing tasks. For the color editing task, it achieves competitive performance. In the material and detail editing tasks, it delivers state-of-the-art results on metrics related to image similarity (CLIP-I, SSIM), demonstrating its ability to effectively capture fine-grained information such as clothing material and neckline details. In the comprehensive editing task, our method exhibits excellent performance, indicating its ability to handle complex editing scenarios involving multifaceted garment attributes.

To comprehensively evaluate the performance of TRAMD-

diff, we further compare it with IP2P, LEDits++, Hive, and TexFit on the DFMM-SpotLight benchmark, as shown in Table IV. We observe that although the general-domain models—IP2P, LEDits++, and Hive—achieve high CLIP-T scores, their CLIP-I, LPIPS, and SSIM scores are significantly worse than those of the fashion-domain models (TexFit and ours). This suggests that while these general-domain models can follow instructions and produce semantically correct edits, they struggle to maintain accurate edit localization and preserve fine-grained fashion details. For the fashion-domain models, our method surpasses TexFit not only in CLIP-T, which measures text-image alignment, but also in CLIP-I, LPIPS, and SSIM, which assess image similarity and structural consistency. These results demonstrate that our model can effectively handle diverse fashion images and editing instructions while maintaining high editing quality.

3) *User Study*: Since the text-guided image editing task lacks a ground truth, using metrics to evaluate the model has certain limitations. To more comprehensively assess the performance of different editing models, we employed a user study with 30 participants. Each participant was required to evaluate 20 pairs of images generated by four different baselines (IP2P, LEDits++, HIVE, and Ours), considering both text alignment and image fidelity in their selections. To avoid potential bias, the source method of each image was not disclosed to the participants. ¹ The preference for

¹<https://v.wjx.cn/vm/ObkVefi.aspx>

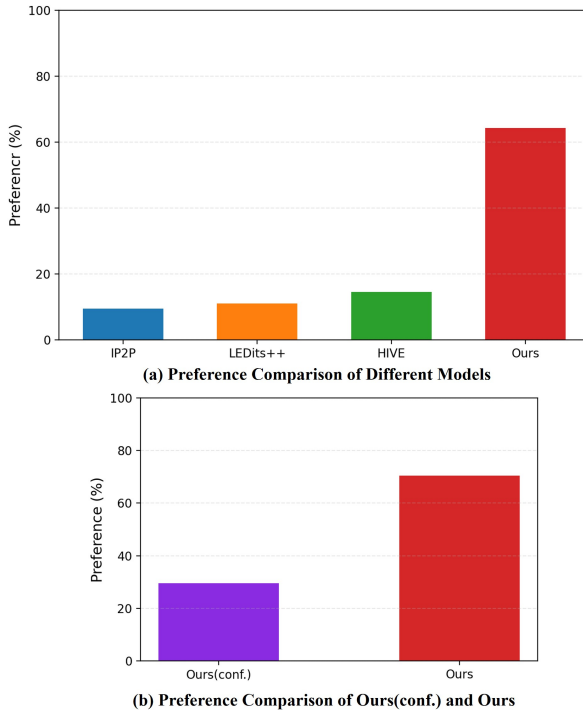


Fig. 10. User Study comparison with State-of-the-Art methods on Fashion-E

images generated by each model is shown in Fig. 10 (a). As seen, our model received the most user preference, surpassing 60%. This demonstrates that our model aligns better with human preferences in both text alignment and preservation of original details, significantly outperforming existing models. In addition, since our model is an improvement based on MADiff (the conference version of our model), we also conducted a separate user study comparing MADiff and our model to validate the enhancements achieved by our approach. As shown in Fig. 10 (b), compared to MADiff, our model achieved over 65% user preference, indicating a significant improvements over MADiff.

C. Ablation Study

1) *Mask Prediction Module (MPM)*: To validate the effectiveness of the Mask Prediction Module (MPM) in improving the accuracy of editing region localization, we conduct an ablation study in which the MPM is removed and the mask is instead derived from the cross-attention map, following the approach used in P2P [7]. As shown in the second row of TABLE V, substituting the MPM with a cross-attention-based mask results in a significant decrease in both CLIP-I and LPIPS scores. This performance drop indicates that the proposed MPM contributes to better preservation of original visual content by more precisely identifying the regions that require editing.

2) *Attention Processor (AP)*: To verify its effectiveness on improving editing magnitude, the Attention Processor is ablated and the inversion noise map is directly used in DDIM sampling. The quantitative results on Fashion-E are shown in Tab. V. Without the Attention Processor, the CLIP-T score notably declines, indicating that simply applying inversion

TABLE V
ABLATION STUDIES ON MPM, ATTENTION PROCESSOR (AP) AND SVIM.

MPM	AP	SVIM	CLIP-T ($\uparrow$)	CLIP-I ($\uparrow$)	LPIPS ($\downarrow$)	SSIM($\uparrow$)
			27.45	84.35	0.168	0.831
✓			27.37	89.53	0.131	0.883
	✓		28.88	88.82	0.157	0.851
✓	✓		29.44	90.20	0.138	0.880
✓	✓	✓	29.36	91.40	0.126	0.895

noise map in DDIM sampling cannot edit the original image to align with the target text. While the Attention Processor proves instrumental in enhancing the editing magnitude.

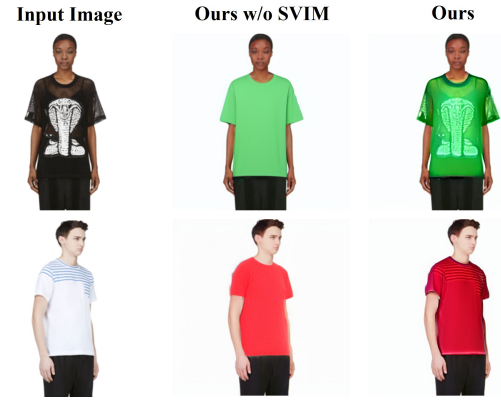


Fig. 11. Visualization results of Self Value Injection Module (SVIM) ablation

3) *Self Value Injection Module (SVIM)*: Since the computation process of the Self Value Injection Module requires the involvement of the mask, we conduct the ablation by comparing the results of MPM + Attention Processor with those of MPM + Attention Processor + SVIM. As shown in Fig. 11, the images generated by the model with the addition of SVIM are more similar to the original images, especially in terms of patterns and textures. This can also be indicated in the third and fourth lines of Tab. V. After adding the SVIM module, the CLIP-I Score significantly increased, while the LPIPS decreased noticeably. This demonstrates that SVIM helps the model retain more information from the original image, particularly high-frequency details.

4) *Hyperparameters*: The ablation experiments of hyperparameters are based on 520 randomly selected samples from Fashion-E, balanced across all task types and the results are presented in Table VI. In Eq. 9, the Gaussian blur in SVIM involves a hyperparameter σ , i.e., standard deviation, which controls the extent of low-frequency information filtered out during high-pass filtering. A larger σ increases the blurriness of the Gaussian blur, thereby preserving more high-frequency components after high-pass filtering. This trend is consistent with our observed metric changes, as shown in rows 1 to 5 of Table VI: as σ increases, CLIP-I tends to improve, indicating that the generated images better retain information from the original images. However, increasing σ may also amplify high-frequency noise, which can slightly reduce the alignment between the target prompt and the edited image (reflected by a slight decrease in CLIP-T). To balance the introduction of high-frequency information and alignment with

TABLE VI
ABLATION STUDIES ON HYPERPARAMETERS.

Threshold	σ	high-pass	CLIP-T ($\uparrow$)	CLIP-I ($\uparrow$)	LPIPS ($\downarrow$)	SSIM($\uparrow$)
0.5N	0.5	Gaussian	29.02	91.52	0.090	0.897
0.5N	1.5	Gaussian	28.87	92.22	0.083	0.904
0.5N	2.5	Gaussian	28.64	92.45	0.080	0.907
0.5N	5.0	Gaussian	28.51	92.66	0.078	0.908
0.5N	7.5	Gaussian	28.50	92.63	0.078	0.909
0.1N	1.5	Gaussian	28.32	91.88	0.085	0.903
0.3N	1.5	Gaussian	28.74	91.99	0.084	0.904
0.5N	1.5	Gaussian	28.78	92.22	0.083	0.904
0.7N	1.5	Gaussian	28.74	91.90	0.083	0.904
0.9N	1.5	Gaussian	28.42	91.92	0.083	0.903
0.5N	1.5	Gaussian	28.78	92.22	0.083	0.904
0.5N	1.5	Laplacian	28.78	92.12	0.084	0.903
0.5N	1.5	Sobel	28.52	90.24	0.096	0.893

the target prompt, we ultimately select a moderate value of 1.5. At this setting, our model maintains relatively high CLIP-T scores while ensuring robust incorporation of high-frequency information, leading to significant improvements in metrics such as CLIP-I.

We further validated the threshold in the attention processor, which determines the proportion of pixels selected based on the top attention map values. As shown in rows 6 to 11 of Table 4, when the threshold is between 0.3N and 0.7N, the model maintains satisfactory editing performance, with the best results achieved at 0.5N. If the threshold is too large (e.g., 0.9N), some pixels in the noise map x_S^{no} with relatively low attention values—i.e., less correlated with the target prompt—will be replaced into the refined noise map x_S^{re} , leading to performance degradation. Conversely, if the threshold is too small (e.g., 0.1N), only a few pixels are introduced into x_S^{re} , preventing the attention processor from functioning effectively. Thus, we set 0.5N as the final threshold for the attention processor.

In the high-pass filtering process of SVIM, we first obtain low-frequency components via Gaussian blur, then subtract these low-frequency components from the original image to derive high-frequency information, thereby achieving high-pass filtering. In the ablation study, we further compared our Gaussian filtering method with other high-pass filtering methods, i.e., Laplacian and Sobel filters. Among these three, the Gaussian blur filter yields the best editing performance, the Laplacian filter performs slightly worse, and the Sobel filter causes a substantial drop in editing capability. Compared to the other two filters, the Sobel filter is highly sensitive to noise, making it prone to introducing high-frequency information with significant noise into the editing process, which adversely affects the final performance. Considering both the superior performance of the Gaussian blur filter and the convenience of controlling filtering strength via its parameter σ , we adopt Gaussian filtering for the value matrix in the final model.

5) *Interpolation strategy*: To evaluate the behavior of our method under various interpolation strategies—where the attention map A is interpolated to 64×64 to resolve resolution discrepancies—we conducted ablation studies using 520 randomly selected samples from Fashion-E, balanced across all task types and tested with different interpolation methods

TABLE VII
ABLATION STUDIES ON DIFFERENT INITIALIZATION (SEED) AND INTERPOLATION STRATEGY.

Interpolation	Seed	CLIP-T ($\uparrow$)	CLIP-I ($\uparrow$)	LPIPS ($\downarrow$)	SSIM($\uparrow$)
nearest	105	28.74	90.09	0.083	0.904
bilinear	105	28.74	92.06	0.083	0.904
bicubic	105	28.78	92.22	0.083	0.904
bicubic	100	28.72	92.26	0.083	0.904
bicubic	95	28.66	92.13	0.083	0.904
bicubic	90	28.83	92.18	0.083	0.904

under the same interpolation seed. The results are presented in the first to third rows of Table VII. The findings indicate that our performance metrics remain consistent regardless of the interpolation strategy employed. This demonstrates that our method is robust across varying interpolation schemes and does not rely on a fixed interpolation strategy.

6) *Different initialization*: For assessing the performance of our method under different initialization conditions, we carried out ablation studies on distinct random seeds for initialization (i.e., different random seeds), with results shown in the fourth to sixth rows of Table VII. The results reveal that our performance metrics remain consistent despite variations in initial conditions. This illustrates that the method exhibits strong adaptability to initialization differences and is not constrained by specific initialization schemes.

TABLE VIII
ABLATION STUDIES ON FILTERED MATRIX.

filtered matrix	CLIP-T ($\uparrow$)	CLIP-I ($\uparrow$)	LPIPS ($\downarrow$)	SSIM($\uparrow$)
K and V (no filter)	25.17	89.80	0.068	0.882
K and V (with filter)	28.79	92.21	0.083	0.904
K (with filter)	25.15	90.06	0.064	0.886
V (with filter)	28.78	92.22	0.083	0.904

7) *Matrix for High-pass Filter*: We also conduct an ablation study base on the same 520 samples on the matrix used for high-pass filtering, with results presented in Table VIII. The first row involves directly replacing the key and value matrices from DDIM inversion with those originally used in DDIM sampling (consistent with the work of MasaCtrl). The second row entails extracting high-frequency information from both the key and value matrices during DDIM inversion and fusing it into the corresponding key and value matrices during DDIM sampling. The third row is extracting high-frequency information from the key matrix during DDIM inversion and fusing it into the corresponding key matrix during DDIM sampling. The fourth row is extracting high-frequency information from the value matrix during DDIM inversion and fusing it into the corresponding value matrix during DDIM sampling. We observe that injecting high-frequency information solely into the value matrix yields results comparable to injecting it into both the key and value matrices. In contrast, injecting high-frequency information only into the key matrix leads to a significant degradation in editing performance. This confirms that high-frequency information indeed resides in the value matrix rather than the key matrix.

V. DISCUSSION

1) *Distinction with higher-resolution inversion:* The reasons why we inject value information instead of using higher-resolution inversion are as follows: higher-resolution inversion demands greater computational resources; furthermore, even with higher-resolution inversion, the texture information of the input image during DDIM sampling cannot be explicitly preserved. In contrast, SVIM reinjects high-frequency information extracted from the original image, effectively preserving texture while enhancing editing strength.

TABLE IX
COMPARISON OF TRAINING PARAMETERS, INFERENCE TIME, AND MEMORY USAGE.

Model	Training Params	Inference Time	GPU memory usage
IP2P	860M	10.53s	18.6GB
HIVE	860M	10.56s	18.6GB
Ours(w/o LLM)	150M	11.13s	15.9GB
Ours(w/ LLM)	150M	11.31s	21.0GB

2) *Training parameters and inference cost:* Based on a 512×512 inference resolution, float32 precision, and an RTX 3090 GPU, the training parameters, inference time, and GPU memory usage of our model are reported in Table IX. As shown, our model uses only about one-sixth of the training parameters of IP2P and HIVE. This enables, allowing efficient training on consumer-grade GPUs while achieving better performance than IP2P and HIVE. Excluding the LLM, it requires less GPU memory than IP2P and HIVE, enabling inference on 16 GB GPUs. And with reduced-precision formats (float16 or int8), inference on 8 GB GPUs is feasible. Including the LLM, total memory usage reaches 21 GB, still fitting on an RTX 3090 and allowing fast inference (11 s per image).

Moreover, since the diffusion-based generation component in our method is training-free, our approach can be applied to any diffusion model. In the future, as diffusion-based base models become sufficiently lightweight, applying our method to more lightweight diffusion models will hold promise for deployment in practical scenarios such as mobile deployment.

3) *Limitations and future plans:* As shown in Fig. 12, our model exhibits two potential limitations.

Firstly, in detail editing tasks, the generated results are sometimes not fully natural; for instance, when converting short sleeves to long sleeves, the sharp edges of the original short sleeves may persist, as shown in the first row of Fig. 12. This issue is also commonly observed in other existing models. Although the problem has been alleviated by the design of our Mask Prediction Module, it still persists. The reason is that our Mask Prediction Module is trained using only the original image, original mask, and target prompt, without generating a target mask corresponding to the edited image based on the target prompt. This can lead to insufficient mask accuracy, resulting in the retention of details from the original clothing in the mask.

Secondly, similar to other training-free model like P2P and PnP, our model struggles to fully handle certain particularly complex textures, as shown in the second row of Fig. 12. This limitation is mainly due to the fact that our diffusion model



Fig. 12. Some failure cases of our method. The issues mainly concern the naturalness of detail editing and the generation of complex textures.

is training-free, and therefore does not internally model the knowledge required for these more intricate textures.

To address these two issues, we plan to construct a more comprehensive dataset in future work, including diverse fashion images, original prompts, target prompts, and masks generated based on the target prompts. This dataset will be used to train both the Mask Prediction Module and the UNet in the Attention-Enhanced Diffusion model, aiming to further mitigate the aforementioned limitations.

VI. CONCLUSION

This paper presents a novel text-guided fashion image editing framework, TRAMDiff, which consists of two key components: the Mask Prediction Module (MPM) and the Attention-Enhanced Diffusion Model. The MPM employs a lightweight U-Net to predict the initial editing region and refines it through a Mask Processor into an addition mask and a removal mask, effectively addressing the issue of inaccurate localization. The Attention-Enhanced Diffusion Model introduces an Attention Processor that replaces the original pixels within the editing region with those of higher attention value, thereby enhancing the editing magnitude. The SVIM, within the Attention-Enhanced Diffusion, incorporates high-frequency information obtained during the inversion process into the sampling stage, substantially improving the preservation of textures and fine-grained visual patterns. Quantitative and qualitative results on the Fashion-E benchmark and the DFMM-SpotLight benchmark, along with a user study, demonstrate that TRAMDiff outperforms existing text-guided editing methods in both text-image alignment and the preservation of original image content.

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